

Performance Testing

Date	15 March 2026
Team ID	xxxxxx
Project Name	AI Smart Patient Appointment & Medication Reminder System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	API Performance Testing & Load Testing
Target Module / API	User Login API, Appointment API, Medication Reminder API, AI Summary API
Test Environment	GitHub Pages (Frontend) + Google Apps Script (Backend) + Google Sheets (Cloud Database)
Test Date	28-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Authentication	10	30	User login completed successfully
2	Appointment Creation	15	45	Appointment stored successfully in Google Sheets

3	Medication Reminde	10	30	Medication saved and SMS reminder triggered
4	AI Healthcare Summary Generation	5	60	AI-generated patient summary returned successfully



Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.15 sec	pass	Fast response
2	Response Time (Max)	< 5 seconds	2.70 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)		24 Req/sec	Pass	Stable request handling
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	45%	Pass	Efficient execution
6	Memory Utilization	< 80%	50%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

The RK Health application performed efficiently during performance testing. Core modules including user authentication, appointment management, medication reminders, AI healthcare summary generation, and report generation responded successfully within the expected response time. The Google Apps Script backend maintained stable communication with Google Sheets, Twilio SMS, Google Calendar, and Groq AI APIs without significant delays. The application remained responsive even when multiple requests were processed simultaneously.

Bottlenecks Identified

- AI summary generation occasionally experienced slight delays because of external Groq AI API response time.
- SMS delivery speed depended on the Twilio network and internet connectivity.
- Google Sheets write operations showed minor latency when processing multiple concurrent requests.

Optimization Steps Taken

- Optimized JavaScript modules to reduce redundant API requests.
- Implemented reusable Google Apps Script helper functions for improved backend performance.
- Added input validation before database operations to reduce unnecessary processing.
- Improved frontend responsiveness by separating CSS and JavaScript modules.
- Optimized Google Sheets read/write operations to minimize execution time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

